Day 109 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Use Coordinates to Find Polygon Lengths and Area

Standards: 6.G.A.3, 6.G.A.1

Learning target: I can draw polygons from coordinates and use side lengths to find area.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 23

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01 / Learn

Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths.

Worked example 1

Plot and connect (0,0), (13,0), (13,8), (0,8). Find the side lengths and area.

1. Plot vertices in order and connect them, including the last to the first.

2. Equal x-values make a vertical side; equal y-values make a horizontal side.

3. Horizontal length 13 - (0) = 13

4. vertical length 8 - (0) = 8

5. area = 13 × 8.

6. State the result in the form requested: Sides 13 and 8; area 104 square units.

Answer: Sides 13 and 8; area 104 square units

Worked example 2

A rectangle has vertices (-3, -4), (5, -4), (5, 4), and (-3, 4). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.

1. Plot vertices in order and connect them, including the last to the first.

2. Horizontal length = 3 + 5

3. vertical length = 4 + 4.

4. Multiply 8 × 8 = 64.

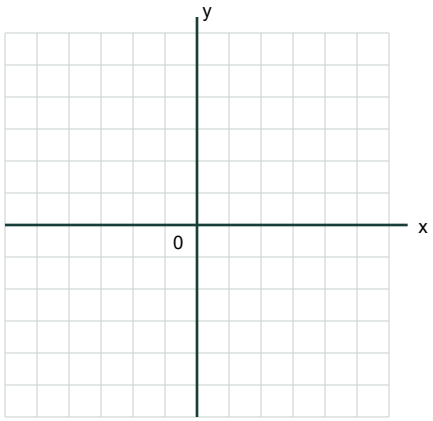
Answer: 8 by 8 units; area 64 square units

Day 109 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

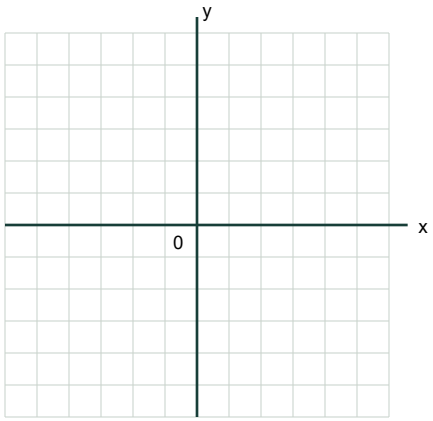
Plot and connect (0,0), (8,0), (8,7), (0,7). Find the side lengths and area.



Label the scale and units.

02 / Constructed response

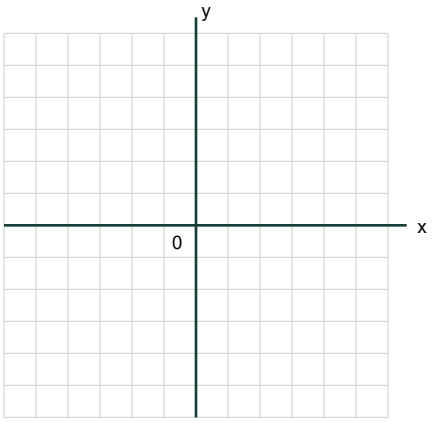
Plot and connect (0,0), (8,0), (8,5), (0,5). Find the side lengths and area.



Label the scale and units.

03 / Constructed response

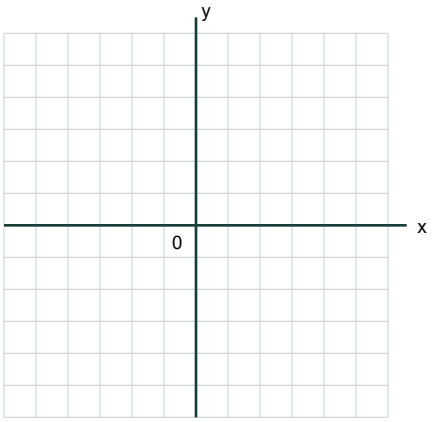
Plot and connect (0,0), (7,0), (7,12), (0,12). Find the side lengths and area.



Label the scale and units.

04 / Constructed response

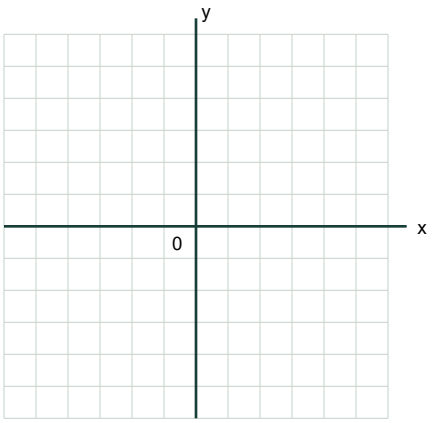
Plot and connect (-3,-5), (6,-5), (6,6), (-3,6). Find the side lengths and area.



Label the scale and units.

05 / Constructed response

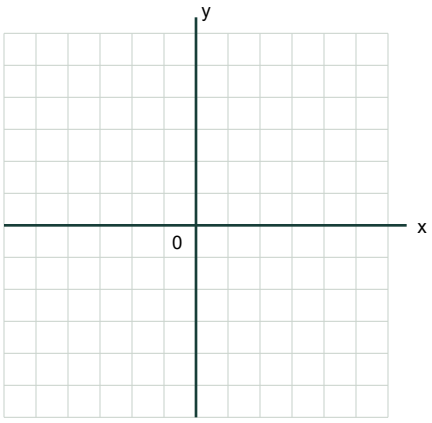
Plot and connect (-4,-2), (2,-2), (2,2), (-4,2). Find the side lengths and area.



Label the scale and units.

06 / Constructed response

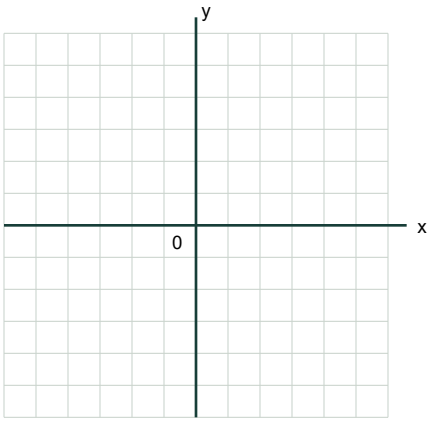
Plot and connect (-3,-4), (2,-4), (2,2), (-3,2). Find the side lengths and area.



Label the scale and units.

07 / Constructed response

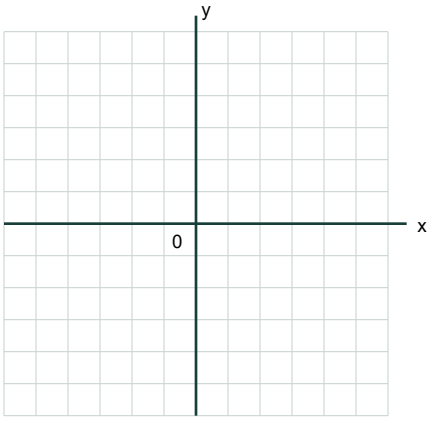
Plot and connect (-8,-3), (6,-3), (6,6), (-8,6). Find the side lengths and area. Find the perimeter too.



Label the scale and units.

08 / Coordinate plotting and two-part response

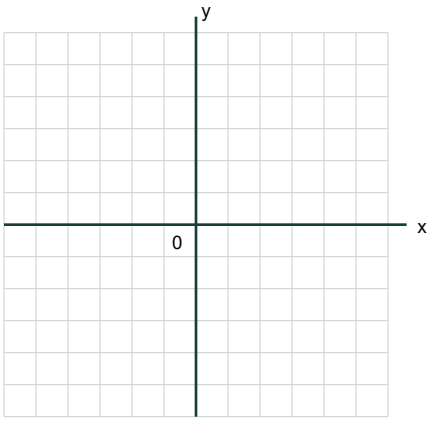
A rectangle has vertices (-6, -4), (4, -4), (4, 4), and (-6, 4). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.



Label the scale and units.

09 / Coordinate plotting and two-part response

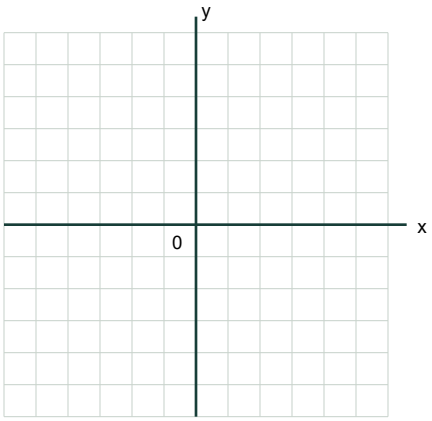
A rectangle has vertices (-5, -4), (5, -4), (5, 4), and (-5, 4). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.



Label the scale and units.

10 / Constructed response

Plot and connect (-7,-3), (4,-3), (4,4), (-7,4). Find the side lengths and area. Find the perimeter too.



Label the scale and units.

Day 109 | Unit 08 | Answer key

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1. Sides 8 and 7; area 56 square units

Horizontal length 8 - (0) = 8; vertical length 7 - (0) = 7; area = 8 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

2. Sides 8 and 5; area 40 square units

Horizontal length 8 - (0) = 8; vertical length 5 - (0) = 5; area = 8 × 5.

6.G.A.3, 6.G.A.1 | Constructed response

3. Sides 7 and 12; area 84 square units

Horizontal length 7 - (0) = 7; vertical length 12 - (0) = 12; area = 7 × 12.

6.G.A.3, 6.G.A.1 | Constructed response

4. Sides 9 and 11; area 99 square units

Horizontal length 6 - (-3) = 9; vertical length 6 - (-5) = 11; area = 9 × 11.

6.G.A.3, 6.G.A.1 | Constructed response

5. Sides 6 and 4; area 24 square units

Horizontal length 2 - (-4) = 6; vertical length 2 - (-2) = 4; area = 6 × 4.

6.G.A.3, 6.G.A.1 | Constructed response

6. Sides 5 and 6; area 30 square units

Horizontal length 2 - (-3) = 5; vertical length 2 - (-4) = 6; area = 5 × 6.

6.G.A.3, 6.G.A.1 | Constructed response

7. Sides 14 and 9; area 126 square units; perimeter 46 units

Horizontal length 6 - (-8) = 14; vertical length 6 - (-3) = 9; area = 14 × 9.

6.G.A.3, 6.G.A.1 | Constructed response

8. 10 by 8 units; area 80 square units

Horizontal length = 6 + 4; vertical length = 4 + 4. Multiply 10 × 8 = 80.

6.G.A.3, 6.G.A.1 | Coordinate plotting and two-part response

9. 10 by 8 units; area 80 square units

Horizontal length = 5 + 5; vertical length = 4 + 4. Multiply 10 × 8 = 80.

6.G.A.3, 6.G.A.1 | Coordinate plotting and two-part response

10. Sides 11 and 7; area 77 square units; perimeter 36 units

Horizontal length 4 - (-7) = 11; vertical length 4 - (-3) = 7; area = 11 × 7.

6.G.A.3, 6.G.A.1 | Constructed response